HEAVY-QUARK SPIN SYMMETRY VIOLATION EFFECTS IN CHARMED BARYON PRODUCTION

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Introduction

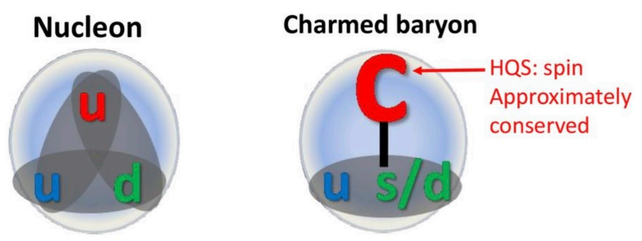


Figure 1: Quark composition of Nucleon and Charmed baryon
Kiyoshi Tanida, 2017

One of the challenging questions in particle physics involves understanding how quarks, the elementary building blocks of matter, and gluons form larger particles known as hadrons through strong interactions. Currently, this fundamental theory is Quantum Chromodynamics (QCD). Studying charmed hadrons is significant because containing a charm quark in heavy hadrons exhibits unique properties, providing a better understanding of QCD. In our work, we study the Conserving & Violating Heavy Quark Spin Symmetry (CHQSS & VHQSS). According to HQSS, it plays a significant role in understanding the dynamics of charm baryons. The consequences of HQSS imply that pseudoscalar and vector D mesons, as well as charm baryon sextets with spins 1/2 and 3/2, form degenerate states[1]. Hence, we incorporate scenarios involving VHQSS to examine its effects on scattering processes and charmed baryon production, Y_c and Y_c', with the exchange of D and D^* mesons, which play a vital role in enhancing the accuracy of our theoretical framework. This is crucial for making precise predictions of production rates at J-PARC, Japan, and in the near future for $\bar{P}ANDA$ at FAIR in Germany.

Objective

We calculate the Conserving Violating contributions of the Heavy Quark Spin Symmetry of the charmed baryons production $p\bar{p} \to Y_c, Y_c'$ processes with the exchanged $D$and D^* meson.

$$p\bar{p} \to \Lambda_c\bar{\Lambda}_c, \Sigma_c\bar{\Sigma}_c, \Sigma_c^*\bar{\Sigma}_c^*, \Sigma_c^*\bar{\Sigma}_c, \Sigma_c\bar{\Lambda}_c, \Sigma_c^*\bar{\Lambda}_c.$$

Theoretical framework

A. $SU(2)_f$ Effective Lagrangians

Firstly, we construct the $SU(2)_f$ effective Lagrangain of D-mesons, nucleon and charmed baryons at chiral power counting Q^0 and Q^1. we obtain

$$\begin{aligned}
\mathcal{L} =& \mathcal{L}^P + \mathcal{L}^A + \mathcal{L}^V + \mathcal{L}^T, \\
\mathcal{L}^P =& g_1\bar{\Sigma}_{ab} i\gamma_5 N_b D_a + g_2\bar{\Lambda} i\gamma_5 N_a D_a + \text{h.c.}, \\
\mathcal{L}^A =& -\frac{g_3}{m_D}\bar{\Sigma}_{ab}\gamma^\mu\gamma_5 N_b\partial_\mu D_a - \frac{g_4}{m_D}\bar{\Lambda}\gamma^\mu\gamma_5 N_a\partial_\mu D_a - \frac{g_5}{m_D}\bar{\Sigma}^\mu_{ab} N_b\partial_\mu D_a + \text{h.c.}, \\
\mathcal{L}^V =& f_1\bar{\Sigma}_{ab}\gamma^\mu N_b D_{\mu a} + f_2\bar{\Lambda}\gamma^\mu N_a D_{\mu a} + f_3\bar{\Sigma}^\mu_{ab}\gamma_5 N_b D_{\mu a} \\
&+ \frac{f_4}{4m_D}\epsilon_{\mu\nu\alpha\beta}\bar{\Sigma}^\mu_{ab}\gamma^\nu N_b D_a^{\alpha\beta} + \text{h.c.}, \\
\mathcal{L}^T =& \frac{h_1}{2m_D}\bar{\Sigma}_{ab}\sigma^{\mu\nu} N_b D_{\mu\nu a} + \frac{h_2}{2m_D}\bar{\Lambda}\sigma^{\mu\nu} N_a D_{\mu\nu a} + \frac{h_3}{2m_D}\bar{\Sigma}^\mu_{ab}\gamma^\nu\gamma_5 N_b D_{\mu\nu a} + \text{h.c.},
\end{aligned}$$

where m_D is D-meson mass ,$D_{\mu\nu} \equiv \partial_\mu D_\nu - \partial_\nu D_\mu$ and $a, b = \{1, 2\}$ are the fundamental indices of the $SU(2)_f$ flavor symmetry. The nucleon N_a, charmed lambda baryon $\Lambda = \Lambda_c$, charmed sigma baryons $\Sigma = \Sigma_c^{ab}$ and D-mesons D_a.

B. Super-multiplet fields in Heavy-Quark Spin Symmetry

As a consequent, the pseudoscalar and vector D mesons as well as spin-$\frac{1}{2}$ and spin-$\frac{3}{2}$ baryons form degenerate states, which can be defined by

$$H_a = \left(\frac{1+v}{2}\right)\left(D_{\mu a,+}\gamma^\mu + i\gamma_5 D_{a,+}\right), \tag{1}$$

$$T^\mu_{ab} = \frac{1}{\sqrt{3}}\left(\gamma^\mu + v^\mu\right) i\gamma_5 \left(\frac{1+v}{2}\right)\Sigma_{(c,+)ab} + \left(\frac{1+v}{2}\right)\Sigma^\mu_{(c,+)ab}, \tag{2}$$

$$T = \frac{1+v}{2}\Lambda_{(c,+)}, \tag{3}$$

These as a building blocks for the HQSS Lagrangian. Taking into account for all of the super-multiplet fields of the C & V HQSS effective Lagrangians.

To construct conserving (C) & violating parts (V). C: $\Gamma^\mu = 1, \gamma_5$, V: $\Gamma^\mu = \gamma^\mu, \gamma^\mu\gamma_5, \sigma^{\mu\nu}$

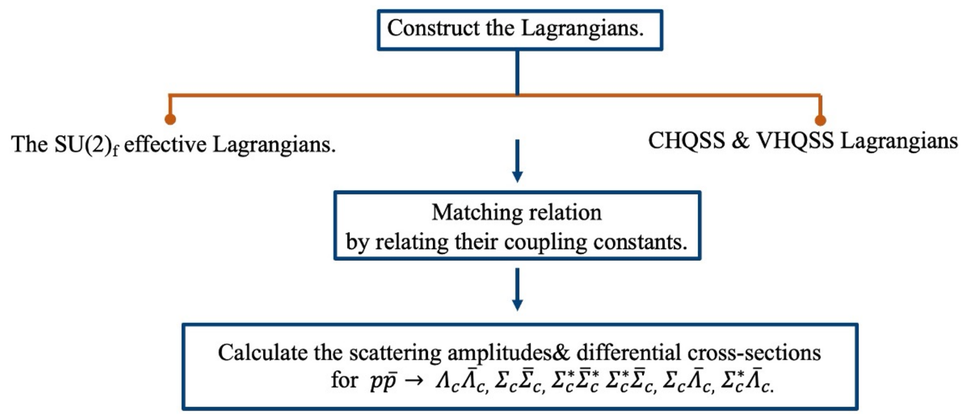
C.The Scattering Amplitudes Differential Cross Section

$$\frac{d\sigma}{dt} = \frac{1}{64\pi(p_{cm})^2 s}\left\langle |M|^2 \right\rangle, \tag{4}$$

$$\mathcal{M}_{p\bar{p}\to Y\bar{Y}'} = \begin{cases} (\sqrt{\alpha}M^{\mathcal{C}}_{p\bar{p}\to Y\bar{Y}'} + \sqrt{\beta}M^{\mathcal{V}}_{p\bar{p}\to Y\bar{Y}'})F(t), \\ (\sqrt{\alpha}M^{\mathcal{C}}_{p\bar{p}\to Y\bar{Y}'} + \sqrt{\beta}M^{\mathcal{V}}_{p\bar{p}\to Y\bar{Y}'})F_n^2(t), n = 1, 2 \end{cases} \tag{5}$$

we include these form factors $F(t), F_n^2(t)$ to express the finite size of hadrons.

Methedology



The relevant coupling constants are given from [2]

Results

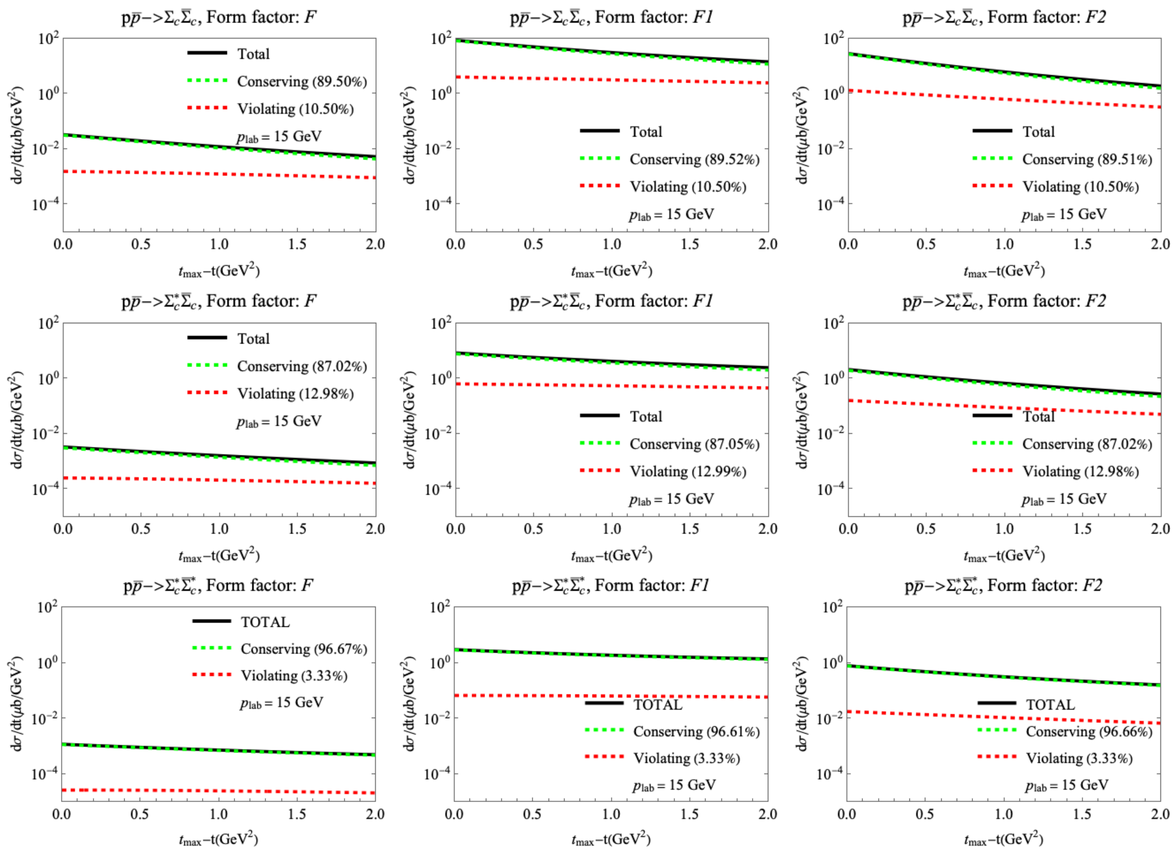


Table 1: The differential cross sections for the reaction $p\bar{p} \to Y_c\bar{Y}_c'$

Discussion

- The differential cross sections of Σ_c and Σ_c^* productions, are mainly contributed by conserving HQSS. The order of magnitudes lies between 10^{-4} and 10^2, as expected. They also depend on the selected set of coupling constants. The contributions with the form factors F_1 and F_2 presented considerable similarity and indicated the protected ranges of the cut-off parameter Λ, suggesting that they could be the cut-off Λ invariant.
- The violating-HQSS contribution significantly shows non-zero, and the quantitative values of approximately 3% are under investigation.
- The calculation of other processes involving Λ_c productions is ongoing.

References

[1] M. F. M. Lutz and M. Soyeur. Radiative and isospin-violating decays of D(s)-mesons in the hadrogenesis conjecture. *Nucl. Phys. A*, 813:14–95, 2008.

[2] T. Sangkhakrit, S.-I. Shim, Y. Yan, and A. Hosaka. Charmed baryon pair production in proton-antiproton collisions in effective Lagrangian and Regge approaches. *Eur. Phys. J. A*, 58(2):32, 2022.